



Deep Learning-Powered Quantum State Estimation

Charlie Low

MPhys Physics
Department of Physics
Durham University

Supervisor: Dr. Will Yeadon

April 20, 2026

Abstract

Quantum state tomography (QST) reconstructs the state of a quantum system from measurement data, and becomes more challenging as the system grows. Classical approaches such as Stokes inversion and Maximum Likelihood Estimation (MLE) are widely used, though both have clear trade-offs. Stokes reconstruction is fast but can produce unphysical states, while MLE guarantees physical validity but has a significantly higher computational cost. This project investigates whether neural networks can provide a useful alternative, with a particular focus on integrating physical structure.

A simulated experimental setup motivated by photonic polarisation tomography was developed to generate noisy measurement data for pure and mixed states. Reconstruction was performed using Stokes inversion, MLE, a baseline fully connected neural network, and a physics-informed neural network (PINN) that encourages physical validity via a Cholesky parametrisation and is trained with a fidelity-based loss. Several convolutional neural network (CNN) architectures were also explored. The methods were compared in terms of reconstruction fidelity, physical validity, and runtime across a range of measurement shot numbers.

The results show that the PINN provides a strong balance of fidelity, physical validity, and computational efficiency. While Stokes reconstruction performs well in low-shot regimes, it frequently produces unphysical states, and MLE remains computationally expensive. The PINN achieves competitive fidelity, maintains strong physical validity, and enables extremely fast reconstruction once trained. Comparisons with CNN-based models suggest that increasing architectural complexity has little effect at higher-shot numbers, and extension to three-qubit systems highlights favourable computational scaling.

Overall, these results indicate that physics-informed neural networks are a promising approach for fast and physically meaningful quantum state tomography.

Contents

1	Introduction and Background	3
1.1	Quantum State Tomography	3
1.2	Physical Motivation	4
1.3	Classical Quantum State Reconstruction	6
1.4	Machine Learning for Quantum State Tomography	7
1.5	Scope and Objectives	8
2	Methodology	9
2.1	Overview of the Reconstruction Pipeline	9
2.2	Quantum State Generation	10
2.3	Tomographic Measurement Model	10
2.4	Noise and Finite Sampling	11
2.5	Dataset Construction	12
2.6	Classical Reconstruction Methods	13
2.7	Baseline Neural Network Model	14
2.8	Physics-Informed Neural Network	16
2.9	Convolutional Neural Network Models	17
2.10	Evaluation Metrics and Statistical Analysis	18
3	Results and Discussion	19
3.1	Classical Baseline Performance	19

3.2	Neural Network Architecture Comparison	20
3.3	Comparison with Classical Methods	23
3.4	Scaling to Three Qubits	27
4	Limitations and Future Work	28
5	Conclusion	29
	Summary for a General Audience	31
	Generative AI Statement	31

1 Introduction and Background

1.1 Quantum State Tomography

Quantum state tomography (QST) is the process of reconstructing the quantum state of a system from measurement data obtained from repeated measurements of identically prepared copies[1, 2]. In general, a quantum state is described by a density matrix ρ , which must satisfy three key physical constraints: it must be Hermitian, positive semidefinite, and have unit trace[3]. These conditions restrict ρ to a highly structured subset of complex matrices representing physically valid quantum states.

The quantum state and experimental observations of it are connected by Born’s rule. For a set of measurement operators $\{E_k\}$, the probability of observing outcome k is

$$p_k = \text{Tr}(\rho E_k). \quad (1)$$

The operators $\{E_k\}$ form a positive operator-valued measure (POVM), which provides the most general mathematical description of a quantum measurement[4]. Each operator E_k is a positive semidefinite matrix associated with a particular measurement outcome, and together they satisfy the completeness relation $\sum_k E_k = \mathbb{I}$. Physically, this ensures that the probabilities of all outcomes sum to one.

POVMs provide a flexible way to describe quantum measurements, including realistic situations where noise, loss, or imperfect detection may be present. A particularly important special case is a projective measurement, where each measurement operator is a projector onto a particular state, and the probability of each outcome is determined by the overlap between the quantum state and that measurement basis. For a projective measurement onto a pure state $|\psi_k\rangle$, this takes the form

$$E_k = |\psi_k\rangle\langle\psi_k|. \quad (2)$$

Projectors therefore represent measurements onto specific basis states, and the corresponding probabilities p_k give the overlap between the state ρ and those measurement directions. In many tomographic settings, especially photonic polarisation tomography, reconstruction is based on these projective measurements, which will be introduced more in Sec. 1.2.

In practice, these probabilities are not known exactly and must be estimated from experimental data. Measurements are performed a finite number of times, with each repetition referred to as a shot. The outcomes of these shots provide a set of counts that can be converted into empirical frequencies, which can be used as estimates of the true probabilities. Due to the inherently probabilistic nature of quantum mechanics, individual measurement outcomes are stochastic. However, as the number of shots increases, the

estimated probabilities converge towards their true values, with statistical uncertainty scaling approximately as $1/\sqrt{N}$.

This highlights an important conceptual point: quantum measurements are intrinsically probabilistic, but increasing the number of shots makes the observed statistics behave more deterministically. As the number of measurements increases, statistical fluctuations are reduced, and the observed frequencies more closely approximate the underlying expectation values.

The goal of QST is to invert this relationship and reconstruct ρ from the measured probabilities. This requires an informationally complete set of measurements, meaning that sufficient independent data is collected to determine all free parameters of the density matrix. For an n -qubit system, the density matrix contains $4^n - 1$ independent real parameters, so the amount of information required grows exponentially with system size[1, 5].

This exponential scaling presents a fundamental challenge for QST. Even for modest numbers of qubits, the size of the state space becomes prohibitively large[2], making both data acquisition and reconstruction increasingly demanding. As a result, developing efficient and scalable reconstruction methods is a central problem and forms a key motivation for the approaches explored in this work.

1.2 Physical Motivation

Although the reconstruction methods developed in this work are evaluated using simulated data, they are motivated by standard photonic QST experiments. Photonic systems provide a useful reference point because their measurement processes are relatively clean and well understood, making them a good setting for developing and comparing reconstruction methods, even when the focus is primarily methodological.

In these systems, qubits are encoded in the polarisation of single photons. The logical basis states are identified with horizontal and vertical polarisations,

$$|H\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |V\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (3)$$

which provide a simple and experimentally accessible description of a two-level quantum system. This description is convenient not only because it is physically intuitive, but also because the associated measurement settings can be implemented directly using standard optical components.

Entangled photon pairs can be generated using spontaneous parametric down-conversion (SPDC)[6], a nonlinear optical process in which a higher-energy pump photon is converted into two lower-energy photons. Under suitable phase-matching conditions, the resulting photon pairs can exhibit strong correlations in their polarisation degrees of freedom, allowing entangled states to be prepared. In practice, the properties of these states depend on the experimental conditions. Short pump pulses tend to produce photons in well-defined temporal modes, leading to highly pure quantum states, while longer pulses introduce temporal distinguishability between photons and therefore lead to more mixed states[7]. This is useful in this context, as it gives a clear physical motivation for analysing both pure and mixed states separately. It also means that the behaviour of the reconstruction methods can later be interpreted in the context of different experimental regimes, with different approaches potentially being more suitable depending on how the states are generated.

A typical measurement setup used for photonic tomography is shown in Fig. 1. Each photon passes through a series of optical elements that define the measurement basis. Quarter-wave plates (QWPs) and half-wave plates (HWPs) are used to rotate the polarisation state, and this is then followed by a polarising beam splitter (PBS) or linear polariser, which performs a projective measurement onto the chosen basis. The outputs are detected using single-photon detectors, and coincidence measurements between the two arms are used to estimate the joint outcome probabilities[6]. This links the experimental setup and the reconstruction problem: the optical elements determine which projectors are measured, and the recorded coincidence measurements provide the data from which the density matrix must be predicted.

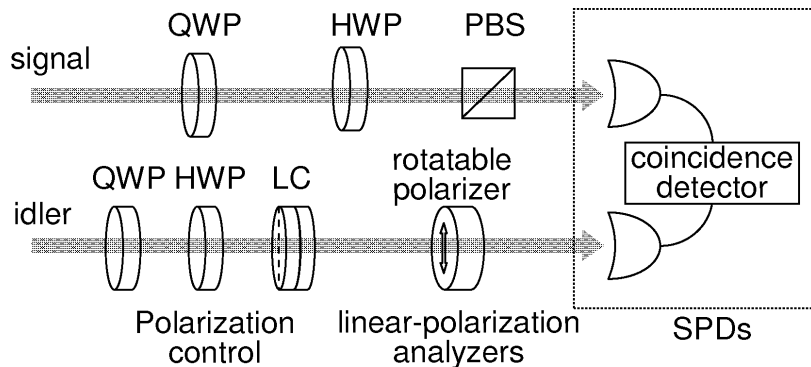


Figure 1: Schematic of a photonic polarisation measurement apparatus, showing waveplate-based basis rotation followed by projective measurement and coincidence detection. Reproduced from Wang and Kanter[6].

By adjusting the waveplate angles, a standard set of six single-qubit measurement states can be accessed. In addition to the $\{|H\rangle, |V\rangle\}$ basis, one also measures in the states

$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \quad |A\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle), \quad (4)$$

$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle), \quad |L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle), \quad (5)$$

corresponding to diagonal/anti-diagonal linear polarisations and right/left circular polarisations respectively. Together with $\{|H\rangle, |V\rangle\}$, these form three mutually unbiased bases, each providing complementary information about the state. Importantly, these bases are not chosen randomly. They are the measurement states naturally available from the optical setup, so the mathematical structure of the tomography problem is directly tied to experimentally accessible measurements[2].

For a two-qubit system, tomography is then built from tensor products of these single-qubit states. This produces a set of $6 \times 6 = 36$ two-qubit projectors, as these projectors span the space of two-qubit operators, allowing all $d^2 - 1 = 15$ independent parameters to be determined[5]. In other words, the probabilities associated with these 36 projectors contain enough information to reconstruct a general two-qubit density matrix. This is an important point for the rest of the project: the input data used throughout is not abstract or arbitrarily constructed, but comes from a physically motivated set of measurements defined by combinations of waveplates, beam splitters, and detectors. That makes it a sensible basis not only for the classical reconstruction methods, but also for the neural network models introduced later.

Real experiments, of course, are not ideal. Several sources of noise can affect the measurement process, including waveplate misalignment, polarisation drift, detector noise, and imperfect state preparation[6]. In the present context, two effects are particularly relevant. The first is waveplate misalignment, where small angle errors cause the implemented measurement basis to differ slightly from the intended one. In the model, this can be interpreted as a small rotation of the measurement operators, and so it provides a natural way to model calibration-type measurement errors. The second is reduced two-photon interference (TPI) visibility, which reflects imperfect indistinguishability or imperfect mode overlap between photons. Experimentally, this is seen as a reduction in the contrast between maximum and minimum coincidence counts in an interference fringe. In practice, reduced visibility washes out correlations and makes the reconstruction problem harder. These particular noise sources connect directly to the quantities being measured, and they therefore provide physically meaningful test cases for comparing reconstruction methods.

At the same time, it is important to keep the scope of this section tight. The aim here is not to model one specific SPDC experiment in full detail, but rather to use a standard photonic tomography setup as a physically motivated reference for defining the measurement operators and noise processes. All results presented in this work are gathered through simulation at the density matrix level, rather than from raw experimental data. This allows controlled and reproducible comparisons between reconstruction methods while still keeping the key features that make practical QST challenging. Although the discussion is grounded in photonic polarisation experiments, the pipeline itself is not restricted to this experiment, and the same ideas carry over to any system for which an informationally complete set of measurements can be defined. The photonic setup should therefore be viewed as a concrete and experimentally meaningful way of motivating the measurement structure used in this project, rather than as a limitation on the broader applicability of the methods.

1.3 Classical Quantum State Reconstruction

Once measurement probabilities have been obtained, the task of QST becomes reconstructing the density matrix ρ from this data. This is a non-trivial inverse problem, and several classical approaches have been developed to perform this reconstruction.

One of the simplest and most widely used methods is linear inversion, often referred to as Stokes reconstruction[8]. In this approach, the measurement probabilities are linearly related to the elements of the density matrix through the known set of projectors. By vectorising the density matrix and expressing the measurement probabilities as a linear system, the reconstruction problem can be written in the form

$$\mathbf{p} = M \text{vec}(\rho), \quad (6)$$

where M is determined by the chosen measurement operators. The density matrix can then be estimated by solving this system, typically using a least-squares approach.

Linear inversion is attractive because it is fast, simple to implement, and does not require any iterative optimisation. However, it has an important limitation: it does not enforce the physical constraints required of a density matrix[1, 9]. In particular, the reconstructed ρ may fail to be positive semidefinite, especially in the presence of noise or limited measurement statistics. This can lead to unphysical states with negative eigenvalues, which do not correspond to valid quantum states, as physical density matrices must be positive semidefinite.

To address this issue, Maximum Likelihood Estimation (MLE) is commonly used[3]. Rather than solving a linear system directly, MLE formulates the reconstruction as an optimisation problem. The goal is to find the density matrix that maximises the likelihood of observing the measured data, subject to the constraints that ρ is Hermitian, positive semidefinite, and has unit trace. This ensures that the reconstructed state is always physically valid[5].

While MLE is a widely used solution to the reconstruction problem, it comes at a significant computational cost. The optimisation is nonlinear and typically requires iterative numerical methods, which must be performed separately for each state. As the system size increases, the dimensionality of the optimisation problem grows rapidly[9], making MLE increasingly expensive in practice[3].

These limitations highlight a central challenge in QST. Linear inversion methods are fast but can produce unphysical results, while optimisation-based methods such as MLE guarantee physicality but are computationally demanding. Both approaches can also be sensitive to noise and experimental imperfections. This trade-off motivates the exploration of alternative reconstruction methods that aim to combine physical validity with computational efficiency, which is the focus of this work.

1.4 Machine Learning for Quantum State Tomography

In recent years, machine learning methods have been explored as an alternative approach to QST[10]. Rather than explicitly solving an inverse problem for each measurement set, these methods aim to learn a direct mapping from measurement data to the underlying quantum state.

Neural networks provide a flexible framework for learning such mappings. At a basic level, a neural network consists of a sequence of layers, each of which applies a linear transformation followed by a non-linear activation function. For an input vector $\mathbf{x}$, a single layer performs the operation

$$\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b}), \quad (7)$$

where W is an $m \times n$ weight matrix that defines a linear transformation from the input vector $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$ to an m -dimensional output space, $\mathbf{b}$ is a bias vector, and σ is a non-linear activation function such as a rectified linear unit (ReLU). This produces an output vector $\mathbf{h}$, which is then passed as input to the next layer. By composing lots of these layers, the network is able to represent highly non-linear functions, and with sufficient width and depth can act as a universal function approximator[11].

The parameters of the network, namely the weights and biases, are learned from data. During training, a dataset of input-output pairs is provided, and the network's predictions are compared to the true outputs using a loss function. This loss quantifies the error of the model, and gradient-based optimisation methods are used to adjust the parameters in order to minimise it. The gradients are computed using backpropagation, which applies the chain rule to propagate error signals backwards through the network. Repeating this process over many iterations allows the network to gradually learn an approximation to the desired mapping.

In the context of QST, the reconstruction problem can be viewed as learning a mapping from measurement data to the underlying quantum state. The input to the network consists of measurement statistics, and the output corresponds to a parametrisation of the density matrix. Once trained, the network can perform this reconstruction in a single

forward pass, replacing the need to solve a separate optimisation problem for each state and shifting the computational cost to an offline training phase. This provides several practical advantages[8]. Inference is extremely fast once training is complete, making neural networks well suited to reconstructing large numbers of states or near real-time applications. They can also learn to account for noise and imperfections in the measurement process[12], and are flexible enough to capture complex non-linear relationships between measurement statistics and quantum states.

However, the mapping from measurement data to quantum states is highly structured and subject to strict physical constraints, which introduces important challenges. A generic neural network does not automatically enforce properties such as positivity and trace normalisation, meaning that naively predicting matrix elements can lead to unphysical states. More generally, standard neural network architectures are often designed for tasks such as image processing, where features like spatial locality are important, which does not naturally match the global and highly structured nature of tomographic data. This shows the importance of incorporating physical structure into the model design. The measurement data arises from a fixed set of known projectors with a clear tensor-product structure, while the space of valid outputs is restricted to physical density matrices. Incorporating this information into the model, whether through the choice of parametrisation, architecture, or loss function, provides a natural way to guide learning and improve both performance and physical consistency.

The central idea explored in this work is therefore not simply to apply neural networks to QST, but to investigate how incorporating physical knowledge into the model influences reconstruction performance. In particular, different architectures and design choices are compared to assess whether more physically informed models offer advantages over more generic approaches.

1.5 Scope and Objectives

The primary aim of this project is to investigate the use of neural network models for QST, with a particular focus on how incorporating physical structure into the model design influences reconstruction performance. The focus is on how the different methods perform in general, using an experimental system as a guide, rather than on optimising performance for a specific physical setup.

To address this, several specific objectives are considered. First, neural network models are constructed to map measurement data directly to quantum states, allowing comparison with the classical reconstruction methods introduced earlier. A key aspect of this is the use of a Cholesky parameterisation of the density matrix, which provides a convenient way to enforce positivity and trace normalisation by construction. This ensures that reconstructed states remain physically valid without requiring additional constraints during optimisation.

In addition to the choice of parametrisation, a different loss function is explored. In particular, a physically motivated loss function based on state fidelity is considered, allowing the training process to directly reflect the quality of quantum state reconstruction. This provides a more physically meaningful optimisation target than standard error losses in the matrix elements, which may not fully capture the relevant physical differences between states.

The influence of model architecture is also investigated. Alongside fully connected neural networks, convolutional neural network (CNN) architectures are explored to assess

whether introducing additional structural assumptions can improve performance. While CNNs are typically associated with spatial data, they provide a way to test whether exploiting structure in the measurement data improves reconstruction.

These elements are combined to evaluate whether more physically informed designs offer advantages over more generic approaches. Throughout, the focus remains on understanding the trade-offs between reconstruction accuracy, physical validity, and computational efficiency.

The remainder of this report is structured as follows. Section 2 describes the construction of the simulated dataset, the implementation of both classical and neural network reconstruction methods, and the evaluation metrics used for comparison. Section 3 then presents the results and accompanying discussion, before concluding with a summary of key findings and potential directions for future work.

2 Methodology

2.1 Overview of the Reconstruction Pipeline

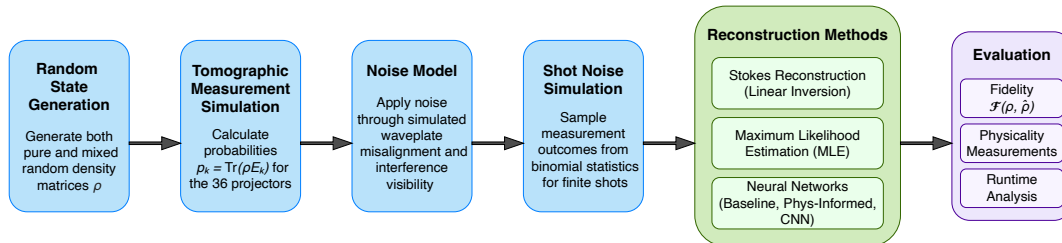


Figure 2: Schematic overview of the reconstruction pipeline used in this work.

The aim of this work is to compare different approaches to QST within a controlled and simulated experimental environment. To allow for a systematic comparison between classical reconstruction algorithms and machine learning methods, a complete reconstruction pipeline for QST was implemented, as shown in Fig. 2.

The pipeline follows the key stages typically encountered in experimental QST. First, random quantum states are generated as the underlying physical states to be reconstructed. Datasets containing both pure and mixed states are generated to explore how the methods perform in different experimental configurations. Tomographic measurements are then simulated by calculating measurement probabilities for an informationally complete set of projectors using Born’s rule. The projectors used in this work are defined by the polarisation bases discussed in Sec. 1.2. These calculated probabilities represent the ideal measurement data for a theoretically perfect experimental setup[8].

However, these probabilities do not fully reflect measurements in a real experiment, so to match realistic experimental conditions more closely, noise models are applied to the measurement operators and finite measurement statistics are simulated through shot noise. This produces measurement data similar to that obtained in real photonic tomography experiments, and ensures that the methods are evaluated under realistic conditions.

The simulated measurement data is then used to reconstruct the density matrix using several reconstruction methods. These include classical approaches, such as Stokes reconstruction and MLE, as well as a range of neural network models trained to learn the mapping between measurement outcomes and density matrix parameters. All methods are applied to the same test datasets to ensure they are evaluated under identical

conditions.

Finally, the reconstructed states are compared to the true states using fidelity-based metrics, to determine their accuracy. Their ability to produce physically valid states as well as the runtimes of the methods are also assessed.

2.2 Quantum State Generation

To evaluate the performance of the reconstruction methods across a broad region of quantum state space, synthetic datasets of random quantum states were generated. All states are represented by density matrices ρ (4×4 for a 2-qubit system), which must satisfy the physical constraints of Hermiticity, positive semidefiniteness, and unit trace.

Pure quantum states were generated by sampling random complex state vectors $|\psi\rangle$ with components drawn from a complex normal distribution and constructing the corresponding density matrix

$$\rho = |\psi\rangle\langle\psi|. \quad (8)$$

This procedure produces random pure states[13] distributed across the state space, corresponding to idealised experimental conditions. The pure states were then mixed with the trace-normalised identity matrix via

$$\rho = (1 - \epsilon) |\psi\rangle\langle\psi| + \epsilon \frac{I}{d}, \quad (9)$$

where $\epsilon \ll 1$ and $d = 4$ is the dimension of the Hilbert space for a 2-qubit system.

Mixed states were generated using the Ginibre ensemble[14], a standard method for producing random density matrices. A random complex matrix G is first constructed with elements drawn independently from a complex normal distribution. A valid density matrix is then obtained through

$$\rho = \frac{GG^\dagger}{\text{Tr}(GG^\dagger)}. \quad (10)$$

This construction guarantees that the resulting matrices are positive semidefinite and trace normalised. The Ginibre ensemble produces mixed states with a range of purities, allowing reconstruction methods to be compared across different regions of state space.

Separate datasets of pure and mixed states were constructed in order to investigate how reconstruction performance depends on the underlying state purity across the different methods. This distinction is motivated by photonic experiments, where ideal state preparation can produce highly pure states and experimental imperfections can introduce mixedness, as discussed previously in Sec. 1.2. Analysing reconstruction accuracy separately for pure and mixed states also allows the performance of each reconstruction method to be evaluated across different experimental regimes. This provides a more comprehensive evaluation of the tomography approaches considered in this work and how they may be applicable to other scenarios where tomographic reconstruction is required.

2.3 Tomographic Measurement Model

QST reconstructs an unknown density matrix from measurement data obtained using an informationally complete set of measurements. In this work, the measurements correspond

to projective measurements in the polarisation bases of photonic qubits. For a single qubit, six measurement states are used:

$$\{|H\rangle, |V\rangle, |D\rangle, |A\rangle, |R\rangle, |L\rangle\}.$$

These states correspond to three mutually unbiased bases in polarisation space. Measurements in these bases provide sufficient information to reconstruct the state of a single qubit.

For a two-qubit system, these operators are tensor products of the single-qubit projectors. This results in a total of $6^2 = 36$ measurement projectors. Together, these projectors form an informationally complete measurement set sufficient to fully determining a two-qubit density matrix.

The measurements are described by a set of operators $\{E_k\}$ corresponding to the possible measurement outcomes. In this work, the measurement operators correspond to projectors onto the polarisation states, for example

$$E_k = |H\rangle\langle H|. \quad (11)$$

For the two-qubit system being studied here, the measurement operators are tensor products of the single-qubit projectors, such as

$$E_k = |H\rangle\langle H| \otimes |V\rangle\langle V|. \quad (12)$$

Given a density matrix ρ and measurement operator E_k , the probability of obtaining outcome k is determined by Born's rule,

$$p_k = \text{Tr}(\rho E_k). \quad (13)$$

These probabilities represent the ideal measurement statistics of a perfect experiment.

For each simulated quantum state, the probabilities associated with all 36 projectors are calculated. These probabilities form the measurement data used for reconstruction. For machine learning models, the probabilities are arranged as a 6×6 grid corresponding to tensor-product combinations of the single-qubit measurement bases. Fully connected neural networks use a flattened 36-dimensional input vector, while CNNs operate directly on the grid representation.

It is important to note that the spatial arrangement of the 6×6 grid does not correspond to any physical geometry of the measurement apparatus. Instead, it is a helpful ordering of the tensor-product measurement operators that provides a structured representation of the measurement outcomes for the machine learning models.

2.4 Noise and Finite Sampling

The process above now allows for the ideal measurement statistics of each state to be calculated, but it still does not represent the imperfections that would be present in a real experiment. Both systematic measurement imperfections and finite counting statistics are introduced here to allow the reconstruction methods to be evaluated under conditions more representative of practical photonic tomography experiments.

Measurement imperfections were modelled by incorporating them into the measurement operators. As discussed previously, the ideal projector set was constructed from the single-qubit polarisation projectors and their tensor products. A small unitary rotation was then applied to the measurement bases in order to simulate waveplate misalignment. This fixed

offset represents a calibration error that affects all measurement settings equally. It has been shown in the literature that a common waveplate misalignment value is 0.5° [15]. The noisy measurement operators were therefore obtained through

$$E'_k = U E_k U^\dagger, \quad (14)$$

where U is the tensor-product unitary describing a 0.5° rotation applied to the projector bases. Measurement probabilities were then calculated from the rotated projectors using Born’s rule. It is important to note that for Stokes reconstruction (see Sec. 1.3), probabilities generated using the misaligned projectors were reconstructed using a matrix constructed from the ideal projectors. This ensures that the applied noise affects the reconstruction in a physically realistic way, rather than being absorbed into the reconstruction operator.

A second source of noise was incorporated through a finite two-photon interference (TPI) visibility. This visibility quantifies how indistinguishable the photons are when they interfere on a beam splitter. As discussed by *Wang and Kanter*[6], TPI visibility ranges from approximately 90% at room temperature to above 99% at -196°C , depending on experimental conditions. In this work, a TPI visibility of 93% was used to represent a typical value for a moderately cooled, well-balanced experimental setup. This noise is applied to the probabilities obtained from the rotated projectors, by mixing them with a uniform distribution according to

$$p'_k = V p_k + (1 - V) u_k, \quad (15)$$

where V is the visibility and u_k is the uniform distribution over the full measurement set. This models reduced measurement contrast by washing the ideal probabilities towards a uniform distribution.

Finite sampling noise was then introduced by converting the noisy probabilities into measurement counts. For a fixed total number of shots N , counts were sampled from the full probability vector using a multinomial distribution. This assumes that identical copies of the state can be repeatedly prepared[5]. The resulting empirical frequencies therefore fluctuate around the underlying probabilities, with statistical uncertainty that decreases as the number of shots increases approximately as $N^{-1/2}$. These empirical frequencies form the actual input data used by the reconstruction methods, allowing the methods to be tested against more realistic experimental data[2].

The number of shots required to approximate the probabilities, as well as the number that can feasibly be taken, varies across experiments, and as such there is not a standard value of shots taken across experiments. A common technique therefore in many studies is to sweep across a range of shots for each dataset[12]. In this work, reconstructions were evaluated across a shot sweep spanning

$$N \in \{10, 20, 40, 80, 160, 320, 640, 1280\}.$$

This range was chosen to cover both low-shot and high-shot regimes. The shot sweep therefore provides a controlled way to assess how reconstruction fidelity, physicality, and runtime depend on the amount of measurement information available.

2.5 Dataset Construction

Using the state generation and noise modelling procedures described above, synthetic datasets of measurement outcomes were generated for training and evaluating the recon-

struction methods. These measurement frequencies were stored alongside the corresponding true density matrices. Separate datasets were constructed for pure and mixed states, as well as for each shot number N in the shot sweep described previously. This allows investigation of how reconstruction performance depends on the underlying state purity, as well as the effect of measurement statistics on the methods.

For each case, a set of 10,000 random quantum states was generated and converted into simulated measurement data using the tomographic measurement model, noise model, and finite sampling procedure. These datasets were then divided into training and test sets. The training set was used to optimise the neural network parameters, while the test set was kept completely separate and used only for the final evaluation of reconstruction performance across the methods. This was done to ensure that the neural network models were not evaluated on training data, preventing overfitting from inflating their apparent performance.

The final datasets consist of simulated tomographic measurement outcomes corresponding to a range of state purities and measurement shot numbers. These datasets provide a controlled environment for comparing classical reconstruction algorithms and machine learning approaches. Additional datasets were also generated for other uses. A dataset was generated with a 50-50 split between pure and mixed states and 320 shots, and this was used to optimise the neural network hyperparameters (see Sec. 2.7). Finally, more datasets were generated in the same way as described previously, but for a 3-qubit system instead. This is analysed and discussed further in Sec. 3.4.

2.6 Classical Reconstruction Methods

Two classical QST reconstruction methods were used as baseline methods in this work: Stokes reconstruction and MLE. These approaches are widely used in tomography and provide useful reference points for evaluating the performance of the machine learning methods explored later.

Stokes reconstruction is a linear inversion method that uses the measurement operators to reconstruct the density matrix directly from the measured probabilities. The method relies on expressing the tomography problem as a linear system relating the measurement probabilities to the density matrix parameters. Using the measurement operators E_k , the probabilities obtained from the tomography measurements are given by Born's rule, as shown in Eq. 13.

By vectorising the density matrix into a parameter vector $\text{vec}(\rho)$, this relation can be written in matrix form as

$$\mathbf{p} = M \text{vec}(\rho), \quad (16)$$

where $\mathbf{p}$ contains the measurement probabilities and M is the measurement matrix constructed from the projectors. Each row of M corresponds to the vectorised form of a measurement operator,

$$M_k = \text{vec}(E_k), \quad (17)$$

so that the matrix maps density matrix parameters to measurement probabilities.

The density matrix can therefore be reconstructed by finding a least-squares solution to the linear system,

$$\text{vec}(\hat{\rho}) = (M^\dagger M)^{-1} M^\dagger \mathbf{p}. \quad (18)$$

In this work, the Stokes matrix was constructed once from the ideal set of tomography projectors and reused for all reconstructions. For each simulated state, the measurement counts were converted into frequencies and used as the probability vector. The reconstructed vector was then reshaped into matrix form, after which Hermiticity and unit trace were enforced. However, the reconstruction does not guarantee positive semidefiniteness, meaning that the resulting density matrix can become unphysical when noise is present in the measurement data[8]. Stokes reconstruction therefore provides a computationally inexpensive but imperfect classical baseline.

MLE provides an alternative approach that reconstructs the density matrix via optimisation. Instead of directly inverting the measurement equations, MLE searches for the physical density matrix that maximises the likelihood of observing the measured data.

Given measurement counts n_k for each outcome k , and model probabilities $p_k = \text{Tr}(\rho E_k)$, the likelihood of observing the data is

$$\mathcal{L}(\rho) = \prod_k p_k^{n_k}. \quad (19)$$

It is more convenient to minimise the negative log-likelihood,

$$\mathcal{L}_{\text{NLL}} = -\sum_k n_k \log(p_k), \quad (20)$$

which forms the loss function used during optimisation.

To ensure that the reconstructed density matrix remains physical throughout the optimisation, the state was parametrised using a Cholesky-like decomposition as introduced earlier. A lower-triangular matrix τ is constructed from a set of real parameters, and the density matrix is defined as

$$\rho = \frac{\tau^\dagger \tau}{\text{Tr}(\tau^\dagger \tau)}. \quad (21)$$

This parametrisation guarantees that the reconstructed state is Hermitian, positive semidefinite, and trace normalised.

In this work, the likelihood optimisation was implemented using gradient-based optimisation with the Adam optimiser, similarly to the approach described in [3]. Starting from an initial state close to the maximally mixed state, the parameters of τ were iteratively updated in order to minimise the negative log-likelihood for the corresponding density matrix ρ . The optimisation was run for 200 steps for each reconstructed state.

Unlike Stokes reconstruction, MLE always produces physically valid density matrices. However, the optimisation procedure must be repeated independently for every state, making it significantly more computationally expensive than linear inversion[8]. These two classical approaches therefore provide complementary baselines for evaluating the neural network models introduced later. Their comparative performance across different experimental conditions is analysed in Sec. 3.1.

2.7 Baseline Neural Network Model

In order to investigate whether machine learning approaches can overcome the limitations of classical tomography methods, a baseline neural network model was first developed. This model provides a simple data-driven reconstruction approach against which more specialised neural network architectures can later be developed and compared.

The baseline model consists of a fully connected feed-forward neural network trained to approximate the mapping between simulated tomographic measurement outcomes and the corresponding density matrix parameters[4]. Once trained, the network reconstructs quantum states from measurement data through a single forward pass[10], making reconstruction significantly faster than iterative optimisation methods such as MLE.

The input to the neural network consists of the flattened measurement frequencies obtained from the QST simulation. For the two-qubit system considered here, the informationally complete measurement set contains $6^2 = 36$ projectors, resulting in a 36-dimensional input vector corresponding to the measured frequencies of each projector outcome. The output of the network is a vector describing the reconstructed density matrix. In the baseline model, the network predicts all real and imaginary components of the 4×4 density matrix without enforcing physical constraints, resulting in 32 output parameters.

The network was trained using the Adam optimiser and the mean squared error (MSE) loss between the predicted density matrix parameters and the true parameters obtained from the simulated states. The loss function is given by

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|^2, \quad (22)$$

where $\mathbf{y}_i$ represents the true parameter vector for state i , and $\hat{\mathbf{y}}_i$ is the corresponding network prediction. Minimising this loss encourages the network to learn the mapping between measurement outcomes and the underlying density matrix parameters[8].

Several architectural and training hyperparameters influence the performance of the neural network. These include network depth, the number of neurons in each hidden layer, the learning rate of the optimiser, the batch size used during training, and the use of dropout regularisation. Dropout was included to reduce overfitting by randomly deactivating a fraction of neurons during training, encouraging the network to learn more general features of the measurement data rather than memorising the training set[16]. Rectified Linear Unit (ReLU) activations were used throughout the network due to their well-established stability during training.

The neural network hyperparameters were optimised using a separate dataset consisting of simulated measurement data generated from a balanced mixture of pure and mixed states with 320 measurement shots. The dataset was divided into training and validation subsets. The network was trained on the training subset for each hyperparameter configuration, while the validation set was used to evaluate the reconstruction fidelity and guide hyperparameter selection.

Hyperparameter optimisation was performed using a two-stage search strategy. Initial experiments explored a broad range of values for key hyperparameters including network depth, layer width, learning rate, dropout rate, and batch size. Based on the reconstruction fidelity observed on the validation dataset, promising local regions of the parameter space were then explored in greater detail. As the network will perform differently for different datasets, hyperparameters were considered near-optimal when small variations around the chosen values changed the validation loss by less than approximately 5%.

The optimal hyperparameters are shown in Table 1.

During training, particular attention was paid to the balance between learning rate and training duration. Larger learning rates allow faster convergence but can lead to unstable training, while smaller learning rates typically require more training epochs to reach optimal performance. The learning rate therefore determines how aggressively the

Hidden Network Layers	Dropout	Learning Rate	Batch Size
(256, 256, 256)	8%	5×10^{-4}	64

Table 1: The hyperparameters selected for the baseline model by the optimisation process.

network weights are updated during optimisation.

Fig. 3 shows the evolution of the validation loss during training for the final model configuration. The lowest validation loss occurs around the 87th training epoch, indicating the approximate point at which the model begins to overfit to the training data. The baseline model used throughout this work was therefore trained for 100 epochs, to ensure that the model had captured all of the information about the mapping.

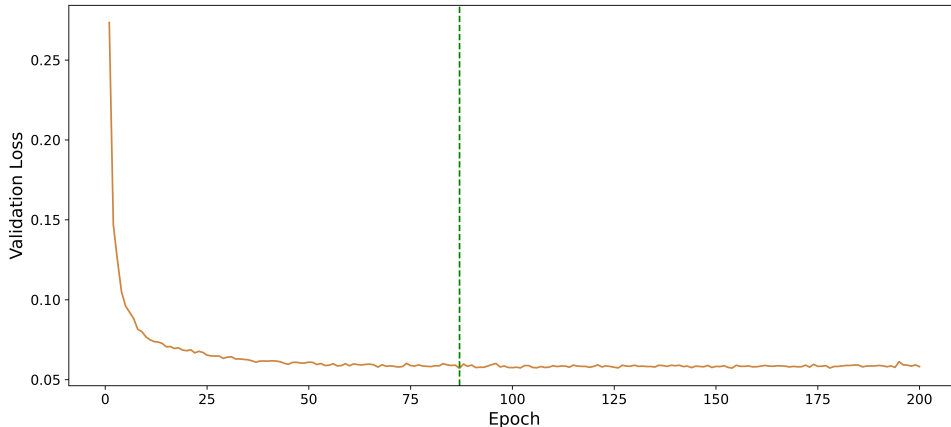


Figure 3: Validation loss during training of the baseline neural network model. The dotted green line indicates the epoch at which the minimum validation loss occurs.

2.8 Physics-Informed Neural Network

While the baseline neural network described previously provides a flexible data-driven approach for quantum state reconstruction, it does not explicitly enforce the physical constraints required of density matrices. In particular, predicting the real and imaginary components of the density matrix directly does not guarantee Hermiticity, positive semidefiniteness, or unit trace. To address this limitation, a physics-informed neural network (PINN) model was developed which incorporates physical structure into both the output parametrisation and the training objective.

The architecture of the PINN remains similar to the baseline network, consisting of a fully connected feed-forward neural network trained on simulated tomographic measurement data. However, instead of predicting the elements of the density matrix directly, the network predicts a set of parameters that are used to construct a valid density matrix through a Cholesky parametrisation.

In this parametrisation, a lower triangular matrix τ is constructed from the network outputs, and the density matrix is defined as

$$\rho = \frac{\tau^\dagger \tau}{\text{Tr}(\tau^\dagger \tau)}. \quad (23)$$

This construction guarantees that the resulting matrix is Hermitian, positive semidefinite, and trace normalised up to numerical precision. By embedding these constraints

directly into the output representation, the neural network is prevented from producing unphysical states during both training and inference. For a d -dimensional quantum system, the matrix τ contains d^2 real parameters corresponding to the degrees of freedom of a complex lower triangular matrix. The neural network therefore predicts these parameters, which are then deterministically converted into the corresponding density matrix using Eq. 23.

In addition to modifying the output parametrisation, the PINN model also uses a loss function based on the Uhlmann fidelity. Although mean squared error provides a stable optimisation objective, it does not directly measure the physical similarity between quantum states. The fidelity between two density matrices ρ and $\hat{\rho}$ is defined as

$$\mathcal{F}(\rho, \hat{\rho}) = \left(\text{Tr} \sqrt{\sqrt{\rho} \hat{\rho} \sqrt{\rho}} \right)^2. \quad (24)$$

Fidelity provides a natural measure of similarity between quantum states, taking values between 0 and 1, where $\mathcal{F} = 1$ corresponds to identical states. The loss function is therefore defined as

$$\mathcal{L}_{\text{fid}} = 1 - \mathcal{F}(\rho, \hat{\rho}), \quad (25)$$

which encourages the network to maximise the fidelity between the predicted and true states during training.

Compared with the baseline model, this physics-informed approach offers two key advantages. First, the Cholesky parametrisation guarantees that all reconstructed states are physically valid density matrices up to numerical precision. Second, the fidelity-based loss directly measures the similarity between quantum states, aligning the training objective with the evaluation metrics used in QST. As a result, the network is encouraged to learn reconstructions that are both physically consistent and optimised for the relevant quantum state similarity measure.

2.9 Convolutional Neural Network Models

While the fully connected neural networks described previously provide a flexible approach to learning the mapping between measurement outcomes and quantum states, they do not explicitly exploit any structure present in the measurement data. In order to better explore whether structured patterns in the measurement data can improve reconstruction performance, CNNs were also investigated as an alternative approach.

In this work, the tomographic measurement outcomes are arranged as a 6×6 grid corresponding to the tensor product combinations of the single-qubit measurement bases. This representation allows the measurement data to be interpreted as a two-dimensional feature map, enabling the use of convolutional architectures.

A standard CNN model was first implemented using this grid representation. Convolutional layers are applied to the input to extract local features, which are then processed through fully connected layers to predict the Cholesky parameters as described in Sec. 2.8. However, the 6×6 grid does not correspond to a physical spatial structure, and neighbouring elements are not necessarily ‘close’ in terms of the underlying measurement operators. A standard CNN therefore imposes an inductive bias that nearby measurement outcomes are correlated, which is not directly motivated by the physical system, as the projectors were arbitrarily arranged in the projector matrix.

To address this limitation, mixing transformations were applied to the measurement data before being input to the CNN. These mixing operators were used to encourage the network to share information between measurement outcomes that represent physically similar projectors, as opposed to those that are simply adjacent in the grid. Given the measurement probability vector $\mathbf{p}$, a transformed input is obtained through

$$\tilde{\mathbf{p}} = \tilde{K}\mathbf{p}, \quad (26)$$

where $\tilde{K}$ is a row-normalised similarity kernel that defines how closely related different projectors are, quantifying how similarly they probe the two-qubit polarisation space. This transformation allows the network to better capture patterns between physically similar projectors.

Two variants of this kernel were explored. The first is an overlap-based kernel,

$$\tilde{K}_{ab}^{(\text{overlap})} = \frac{\text{Tr}(P_a P_b)}{\sum_k \text{Tr}(P_a P_k)}, \quad (27)$$

which produces a relatively smooth mixing of measurement outcomes. The second is a fidelity-inspired kernel,

$$\tilde{K}_{ab}^{(\text{fid})} = \frac{|\text{Tr}(P_a P_b)|^2}{\sum_k |\text{Tr}(P_a P_k)|^2}, \quad (28)$$

which produces a sharper, more localised view of similarity.

In addition to dense mixing, a sparse variant was also explored. In this approach, only the top- k most similar measurement settings for each projector are retained, producing a sparse similarity matrix. This can be interpreted as defining an adjacency-like structure between measurement outcomes, where each setting is connected only to its most similar neighbours. The resulting sparse matrix is then row-normalised and used in the same way as the dense mixing transformation. This encourages the CNN to focus on the most informative relationships, rather than averaging over all measurement settings. In this work, the top 8 most similar settings were used to allow a clear comparison between dense mixing using all 36 projectors, sparse mixing, and the standard CNN where no mixing is applied.

These CNN-based models therefore provide a range of approaches for incorporating physically motivated structure into the reconstruction process, allowing their impact on reconstruction performance to be assessed.

2.10 Evaluation Metrics and Statistical Analysis

The performance of the reconstruction methods was evaluated using a combination of fidelity-based accuracy metrics, physicality checks, and runtime measurements.

The primary measure of reconstruction accuracy is the quantum state fidelity between the true state ρ and the reconstructed state $\hat{\rho}$, defined as

$$\mathcal{F}(\rho, \hat{\rho}) = \left(\text{Tr} \sqrt{\sqrt{\rho} \hat{\rho} \sqrt{\rho}} \right)^2. \quad (29)$$

Fidelity provides a physically meaningful measure of similarity between quantum states, taking values between 0 and 1, where $\mathcal{F} = 1$ corresponds to identical states. For each reconstruction method, the mean fidelity across all reconstructed states was used as the

primary performance metric, with standard error given by $SE = \frac{\sigma}{\sqrt{n}}$, where σ is the standard deviation, and $n = 10,000$ is the number of states averaged across.

In addition to fidelity, the physical validity of the reconstructed density matrices was assessed. In particular, reconstructed states were checked for Hermiticity, positive semidefiniteness, and unit trace. These checks are especially relevant for methods such as Stokes reconstruction, which do not enforce physical constraints by construction and may therefore produce unphysical states in the presence of noise. Ensuring physicality is essential in practice, as reconstructed states must correspond to physically realisable quantum states.

The computational efficiency of each reconstruction method was also evaluated through runtime analysis. The average time required to reconstruct a batch of quantum states was measured for each method, allowing a comparison between the fast inference of neural network models and the iterative optimisation required by methods such as MLE.

To assess whether observed differences in reconstruction performance between neural network models were statistically significant, paired t-tests were performed between the fully connected and the CNN-based neural network models. For each pair of models, the fidelity values obtained on the same test dataset were compared, allowing differences in mean fidelity to be tested while accounting for correlations between samples. This gave a quantitative measure of whether improvements in fidelity are meaningful instead of arising from statistical fluctuations alone.

3 Results and Discussion

3.1 Classical Baseline Performance

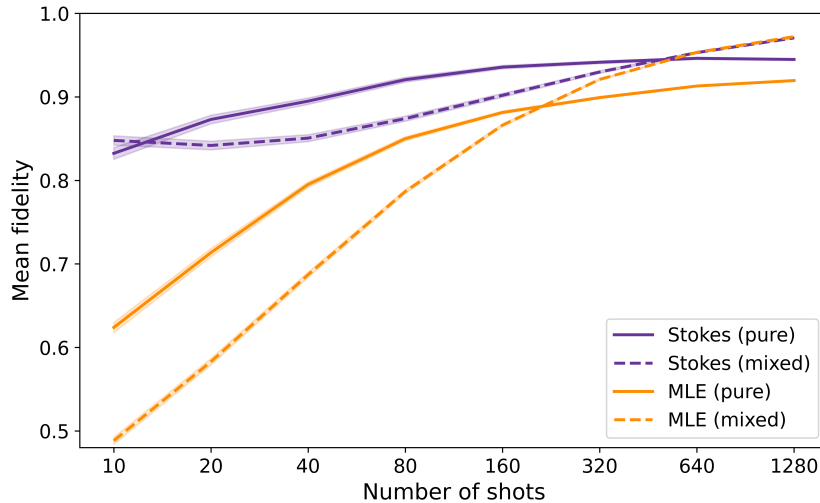


Figure 4: Mean reconstruction fidelity as a function of the number of measurement shots for Stokes inversion and MLE, evaluated on both pure and mixed quantum states. The error sleeves represent the standard error on the mean, but are too small to be visible in the graph.

The performance of the classical reconstruction methods discussed in Sec. 2.6 is shown in Fig. 4 for reconstructing both pure and mixed quantum states in the test sets. These methods provide a baseline against which the neural network approaches can be compared.

For pure states, Stokes reconstruction achieves relatively high fidelity even at low shot numbers, starting at 0.84 for 10 shots and increasing steadily to 0.95 at higher shot counts.

This reflects the efficiency of linear inversion when sufficient measurement information is available.

In contrast, MLE shows significantly lower fidelity at low shot numbers for pure states, beginning at 0.62 for 10 shots. This is expected, as the optimisation procedure is strongly influenced by statistical noise when measurement data is limited. As the number of shots increases, the MLE reconstruction improves substantially, approaching Stokes performance at higher shot counts, although it remains slightly lower overall for pure states.

For mixed states, the behaviour differs slightly. Stokes reconstruction again performs consistently well across all shot numbers, with fidelity increasing smoothly from 0.85 to 0.97. MLE, however, shows a much stronger dependence on shot count. Starting from a lower baseline fidelity of 0.49 at 10 shots, it exhibits a rapid increase in fidelity as more measurement data becomes available, eventually matching Stokes performance at high shot numbers.

This crossover is worth noting. While Stokes reconstruction provides stable performance across all regimes, MLE benefits more significantly from increased data, ultimately yielding similar fidelities once sufficient statistics are available. This can be explained by how the methods operate. Stokes relies on a direct linear inversion of the measurement probabilities, so even when there is a lot of counting noise present, a reasonable reconstruction can still be found. MLE however, relies on minimising over a loss landscape. When counting noise is very large, the landscape becomes much less well-defined, making it harder for the optimisation to converge to a solution that accurately represents the true state. This behaviour is particularly relevant in experimental settings, where low shot regimes are common and reconstruction performance across different shot regimes becomes critical.

Overall, these results illustrate the key trade-offs between the two classical approaches. Stokes reconstruction is fast and performs well even at low shot numbers but lacks guarantees of physicality, as shown later in Sec. 3.3. MLE, while computationally more expensive and initially less accurate in low-shot regimes, provides physically consistent reconstructions and achieves comparable performance when sufficient measurement data is available. These classical methods establish a strong baseline against which the neural network approaches can be evaluated.

3.2 Neural Network Architecture Comparison

The performance of the different neural network architectures is shown in Figs. 5 and 6. These results are used to assess the impact of incorporating physical structure into the neural network model, and to determine whether more complex CNN-based approaches provide a meaningful improvement.

Fig. 5 compares a naive fully connected neural network with the PINN. For pure states, the PINN consistently achieves higher reconstruction fidelity across all shot numbers. For mixed states, the naive model performs slightly better at low shot numbers, before the two methods converge at higher shot numbers. The improvement from the PINN is particularly pronounced for pure states, where the naive model struggles to accurately reconstruct states near the boundary of the state space. This behaviour is consistent with the use of a fidelity-based loss, which is particularly sensitive to errors near the boundary of the state space, corresponding to pure states.

In addition to improved fidelity, the PINN also shows significantly better physicality behaviour. While the naive network frequently produces unphysical density matrices, particularly at higher shot numbers for pure states, the physics-informed approach maintains

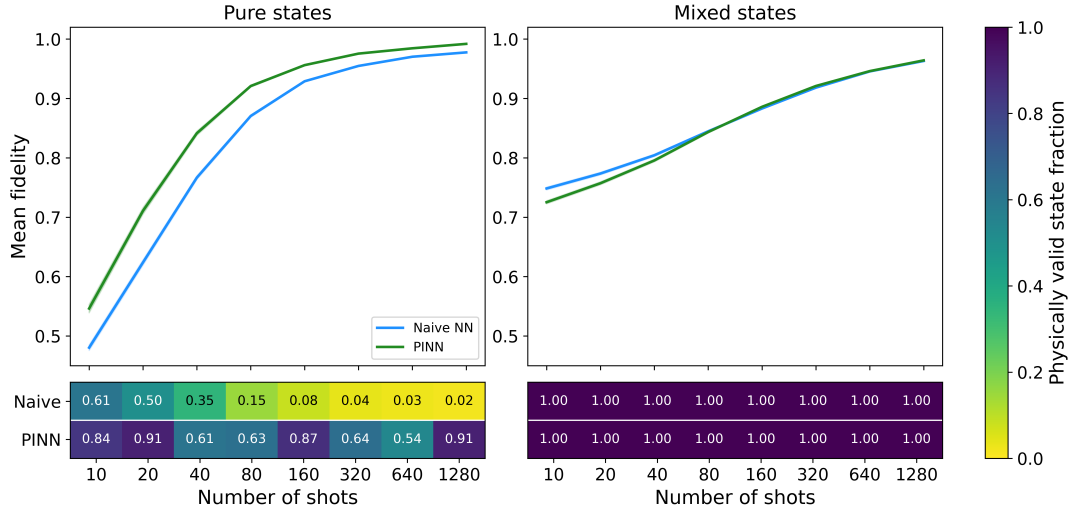


Figure 5: Comparison between a naive fully connected neural network (blue) and the PINN (green). Top panels show mean reconstruction fidelity as a function of the number of measurement shots for pure and mixed states. Bottom panels show the fraction of reconstructed states that are physically valid. Error sleeves represent the standard error on the mean and are too small to be visible.

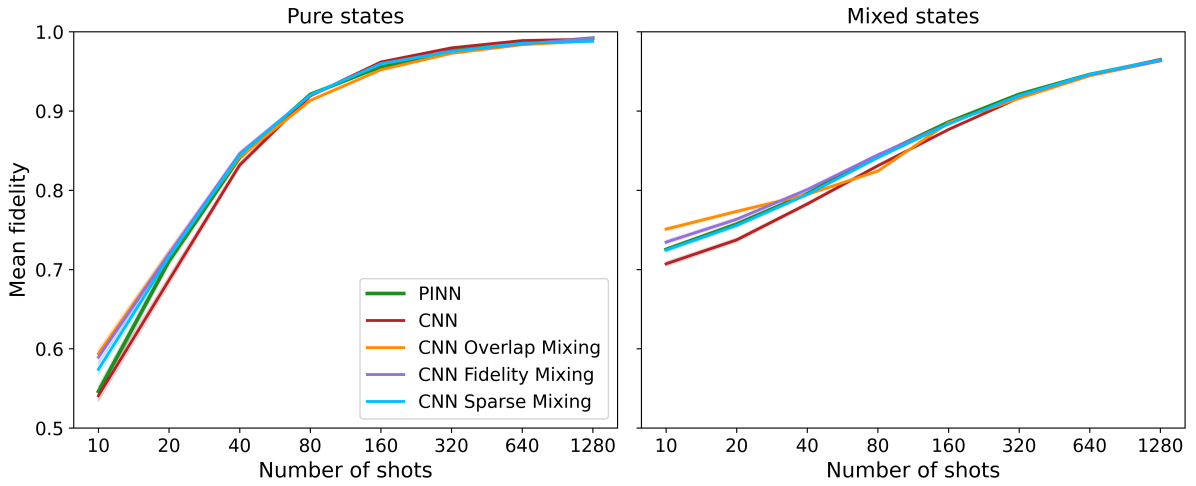


Figure 6: Comparison of the PINN with several CNN-based architectures discussed in Sec. 2.9, including standard CNN, overlap mixing, fidelity mixing, and sparse mixing CNNs. Results are shown for both pure and mixed states as a function of the number of shots. Error sleeves are too small to be visible.

a high fraction of valid states across all regimes. This demonstrates that physics-informed constraints can improve both accuracy and consistency of the reconstruction. It is worth noting, however, that the PINN does not produce perfectly physical states for pure targets in all cases. This arises because pure states lie on the boundary of the state space, and in this regime, small numerical errors in the predicted Cholesky parameters, combined with finite precision when constructing $\rho = \tau^\dagger \tau / \text{Tr}(\tau^\dagger \tau)$, can lead to small violations of positive semidefiniteness, even though the parametrisation enforces physicality in principle.

The comparison with CNN-based models is shown in Fig. 6. All CNN variants achieve performance comparable to the PINN, with very similar fidelity curves for both pure and mixed states. At low shot numbers, some CNN architectures show slightly improved performance, suggesting that convolutional layers may exploit local patterns in the measurement data when the available information is limited. This indicates that CNNs can provide a useful inductive bias in low-data regimes, where the available measurement statistics are limited.

However, as the number of shots increases, the performance of all models converges, and differences between architectures become negligible. This suggests that once sufficient measurement information is available, the additional complexity of CNN-based models does not provide a meaningful advantage over the simpler physics-informed approach. This implies that as the data approaches informational completeness, local correlations between neighbouring measurements in the grid representation become less informative than global relationships across the full measurement set. This is particularly relevant for experimental regimes where high shot numbers are not feasible.

To quantify these observations, paired t-tests were performed comparing the PINN to each CNN variant across all shot numbers in the sweep. The results are summarised in Table 2.

Model Comparison	Mean Difference	t-statistic	p-value
Overlap Mixing CNN vs Phys.-Inf.	3.20×10^{-3}	0.82	0.42
Fidelity Mixing CNN vs Phys.-Inf.	4.73×10^{-3}	1.73	0.10
Sparse Mixing CNN vs Phys.-Inf.	1.57×10^{-3}	0.85	0.41
Standard CNN vs Phys.-Inf.	-6.53×10^{-3}	-2.86	0.012

Table 2: Results of paired t-tests comparing CNN-based architectures with the PINN. Mean difference is defined relative to the PINN.

For the overlap, fidelity, and sparse mixing CNNs, the p-values are all above the standard significance threshold of 0.05, indicating that the observed improvements in fidelity are not statistically significant. This suggests that these architectures do not provide a meaningful performance gain over the PINN.

In contrast, the standard CNN shows a statistically significant decrease in performance relative to the PINN ($p = 0.012$). This provides evidence that a naive convolutional approach, without additional structure or mixing mechanisms, is less effective for this task. However, the magnitude of this difference remains small, indicating that while the effect is measurable, it is small in practical terms.

Overall, these results demonstrate that incorporating physical structure into the neural network provides a clear and consistent improvement over naive architectures. While CNN-based approaches can offer slight benefits in low-shot regimes, these gains are not statistically significant and diminish as more data becomes available. The PINN therefore provides the best balance between performance, simplicity, and interpretability, and is selected as the final model for further analysis.

3.3 Comparison with Classical Methods

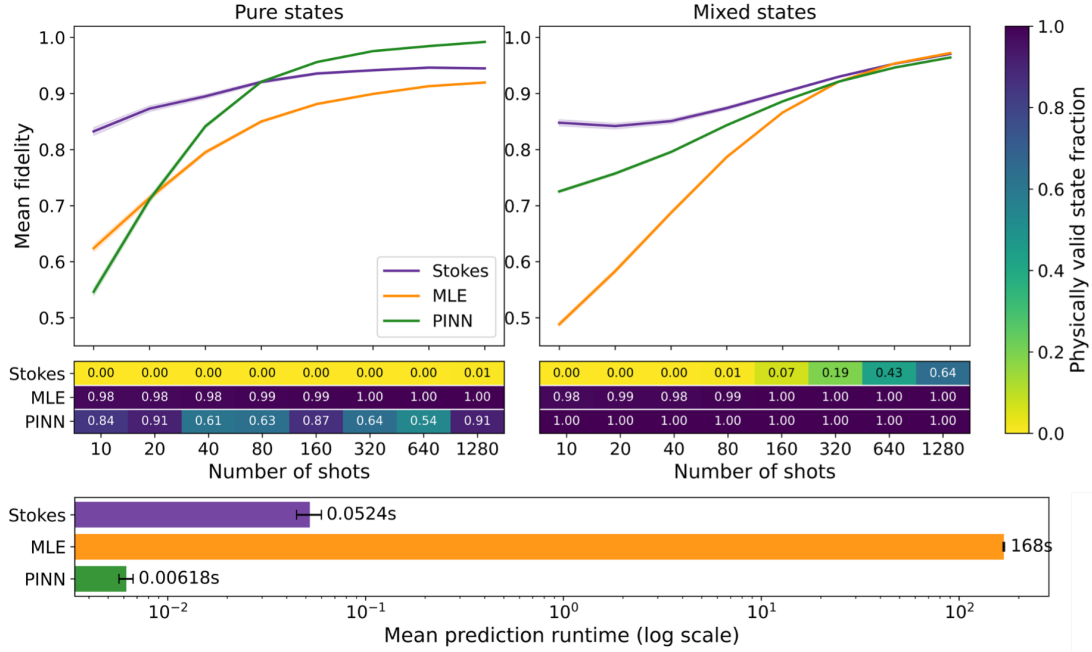


Figure 7: Comparison of Stokes reconstruction, MLE, and the PINN. Top panels show mean reconstruction fidelity as a function of the number of measurement shots, and middle panels show the fraction of physically valid reconstructed states. Pure states are shown on the left, and mixed states are shown on the right. The bottom panel shows mean reconstruction runtime for the 10,000 states on a logarithmic scale. Error sleeves on the fidelity graphs represent the standard error on the mean and are too small to be visible, and the error bars on the runtime graph represent the standard error of the runtime.

Following the selection of the PINN as the final model, its performance can now be compared directly with the classical reconstruction methods, Stokes inversion and MLE. The methods are evaluated using three complementary metrics in Fig. 7: reconstruction fidelity, physicality of the reconstructed states, and computational runtime.

For pure states, the fidelity results show a clear change in the relative performance of the reconstruction methods as the number of shots increases. At very low shot numbers, Stokes reconstruction performs best, followed by MLE and the PINN, initially achieving much lower fidelities than Stokes. This suggests that the neural network and MLE are both more strongly affected by statistical noise when measurement data is scarce, whereas the direct linear inversion used in Stokes reconstruction is able to reasonably estimate states even in a strongly noise-limited regime. This is likely because Stokes treats noise as a simple linear error, rather than attempting to infer the underlying distribution, making it less sensitive to statistical uncertainty in the measured probabilities. As the number of shots increases, however, the neural network improves rapidly, first overtaking MLE and then surpassing Stokes, becoming the best-performing method in the high-shot regime. This behaviour suggests that once enough measurement information is available, the network is able to exploit its architecture to learn a nonlinear inverse mapping that better accounts for the effects of measurement noise. In contrast, Stokes reconstruction simply performs a fixed linear inversion based on an idealised measurement model and cannot adapt to these imperfections. MLE also improves steadily with shot number, but remains below both Stokes and the neural network across the upper-end of the shot range.

Its poorer low-shot performance is consistent with the fact that it relies on iterative optimisation over a likelihood landscape, which becomes less informative when the measured frequencies are strongly affected by statistical noise.

For mixed states, the behaviour is less sharply separated. Stokes reconstruction again begins with the highest fidelity, the neural network starts slightly below it, and MLE gives the lowest initial performance by far. All three methods then improve gradually with increasing shot number and appear to converge towards a common value of around 0.97 in the high-shot regime. This convergence suggests that, for mixed states, the methods are approaching a performance ceiling set not only by finite shot noise but also by the underlying measurement noise present in the simulated data. In other words, once enough shots are available to suppress most of the statistical uncertainty, the limiting factor is no longer sampling error or the reconstruction method used, but the fact that all three methods are reconstructing from noisy measurement data. This behaviour can be understood by noting that mixed states lie in the interior of the state space, where their eigenvalues are more evenly distributed. As a result, small perturbations in the reconstructed density matrix have a relatively minor effect on the overall state, making the reconstruction problem intrinsically more stable than for pure states, which lie on the boundary and are more sensitive to noise. In this regime, the task itself becomes less sensitive to the choice of reconstruction method, and the performance of all approaches is ultimately limited by the same underlying noise. The fact that the curves still appear to trend upwards suggests that fidelities closer to unity could likely be reached at even higher shot numbers, but this regime has not yet been explored in this work. Overall, these fidelity results show that Stokes reconstruction is particularly strong in the low-shot regime, while the PINN becomes increasingly competitive as more information becomes available, eventually giving the best performance for pure states and matching the classical methods for mixed states.

The physicality results in Fig. 7 reveal a clear distinction between the reconstruction methods. Stokes reconstruction produces unphysical density matrices in nearly all cases for both pure and mixed states, with the fraction of valid states remaining effectively zero for pure states and only increasing slightly for mixed states at higher shot numbers. This behaviour arises from the fact that Stokes reconstruction is based on a direct linear inversion of the measurement data and does not enforce the positivity constraint on the density matrix. As a result, even small amounts of noise in the measured probabilities can lead to negative eigenvalues, pushing the reconstructed state outside the physically valid region. The slight increase in physicality observed for mixed states at higher shot numbers is consistent with reduced statistical noise, which brings the reconstructed matrices closer to the true physical state, although without any guarantee of validity.

In contrast, MLE produces physically valid states in all cases for both pure and mixed states. This is expected, as the reconstruction is performed through a constrained optimisation procedure that explicitly enforces positivity and unit trace throughout. As a result, the solution is always a valid density matrix, regardless of the level of noise in the measurement data.

The PINN also shows strong physicality performance, achieving perfect physicality for all mixed states across the full range of shot numbers. However, for pure states, the behaviour is more complex, with the fraction of valid states fluctuating across shot numbers rather than increasing monotonically. This can be understood by considering that pure states lie on the boundary of the state space, where the density matrix is highly sensitive to small perturbations. In this regime, even minor numerical errors in the predicted

Cholesky parameters, combined with finite precision in constructing $\rho = \tau^\dagger \tau / \text{Tr}(\tau^\dagger \tau)$, can lead to slight violations of positive semidefiniteness. This sensitivity explains both the occasional loss of physicality and the non-monotonic behaviour observed as the number of shots increases. Unlike fidelity, which improves steadily as more information becomes available, physicality near the boundary is dominated by numerical stability rather than data quality. As a result, small variations in the network output can lead to fluctuations in whether the reconstructed state remains strictly physical. For mixed states, which lie in the interior of the state space and have more evenly distributed eigenvalues, these effects are not present, and the reconstruction remains robustly physical across all regimes.

The runtime results show a clear separation in computational cost between the three reconstruction methods. MLE is significantly slower than both Stokes reconstruction and the neural network, with an average reconstruction time per set of 10,000 states of 168 s compared to 0.0524 s for Stokes and 0.00618 s for the neural network. This large difference arises from the fact that MLE requires solving a constrained optimisation problem for each individual reconstruction. The method iteratively searches over the space of valid density matrices, repeatedly evaluating the likelihood function until convergence is reached. As a result, the computational cost is high and scales poorly, even in the two-qubit setting considered here. In contrast, Stokes reconstruction is computationally inexpensive, as it involves only a direct linear inversion of the measurement data. This is a deterministic procedure with a fixed computational cost, requiring no iterative refinement, which explains its relatively fast execution time. The neural network achieves the fastest reconstruction time overall, as it predicts states through a single forward pass. This consists primarily of matrix multiplications and activation function evaluations, which are highly optimised operations. It is important to note that the reported neural network runtime corresponds only to prediction time, not training. While training the network incurs a significant computational cost, this is performed offline. Once trained, the model can be reused to reconstruct new states rapidly, shifting the computational burden away from individual reconstructions and into a one-time training process.

Further insight into reconstruction performance is provided by the fidelity distributions shown in Fig. 8. At low shot numbers, the distributions for all methods are broad, reflecting the increased variability when limited measurement information is available. In this regime, the shapes of the distributions reveal clear differences between the methods that are not captured by the mean fidelity alone. For Stokes reconstruction, the distribution is strongly skewed, with a large concentration of states achieving high fidelity and a long tail extending towards lower values. This indicates that the majority of reconstructions are accurate, but a relatively small number of poor estimates reduce the average fidelity. In contrast, the neural network and MLE show wider, more evenly spread distributions at low shot numbers, indicating greater overall variability in their predictions. This is consistent with both methods being more sensitive to statistical noise, leading to a broader range of outcomes rather than a small number of extreme failures. As the number of shots increases, the distributions for all methods become progressively narrower and more concentrated around high fidelity values. This reflects the reduction in statistical uncertainty and the increasing reliability of the reconstructions. The neural network shows a particularly strong reduction in spread, with the distribution tightening more rapidly than MLE, indicating improved robustness as more measurement information becomes available. By the highest shot numbers, all methods have narrow, sharply peaked distributions, consistent with the convergence observed in the mean fidelity results. At the highest shot number, the mixed state distributions for all methods are nearly identical

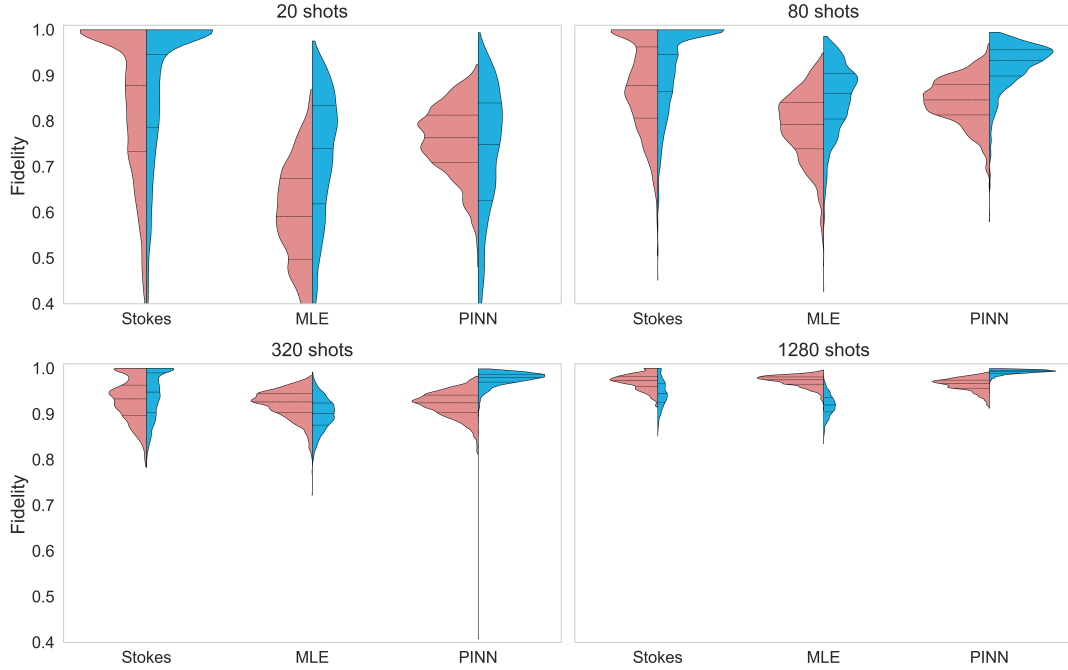


Figure 8: Distribution of reconstruction fidelities for Stokes, MLE, and the PINN at shot numbers (20, 80, 320, and 1280). Distributions are shown separately for pure (blue) and mixed (red) states, with central lines indicating median and interquartile range. The y-axis has been cut at a fidelity of 0.4 for clarity in the graphs.

and tightly concentrated, whereas for pure states a small residual spread remains for both Stokes and MLE, while the neural network shows a more tightly peaked distribution, indicating slightly more consistent performance in this regime. Overall, these distributions show that differences between methods at low shot numbers are driven primarily by variability rather than average performance. While Stokes reconstruction achieves high fidelity for most states but suffers from occasional large errors, the neural network and MLE exhibit more uniformly distributed uncertainty. As the number of shots increases, these differences become smaller, and the data becomes sufficiently informative that all methods produce consistently high fidelity results.

Taken together, these results highlight that no single reconstruction method is optimal across all metrics, with each approach exhibiting distinct strengths and limitations. Stokes reconstruction provides strong fidelity, particularly in the low-shot regime, and is computationally inexpensive, but fails to produce physically valid states. MLE guarantees physicality and achieves high fidelity as the number of shots increases, but does so at a significant computational cost due to the need to solve a constrained optimisation problem for each reconstruction. The PINN does not outperform the classical methods in every regime, but is the only approach in this study that combines competitive fidelity, strong physicality, and extremely fast reconstruction once trained.

This balance makes the neural network particularly well suited to practical tomography scenarios where many reconstructions must be performed efficiently. While Stokes reconstruction remains useful for fast, approximate estimates and MLE for high-confidence, physically constrained offline analysis, the neural network provides a flexible middle ground that is both reliable and scalable. The fact that this performance is achieved through a physics-informed architecture further suggests that incorporating prior knowl-

edge of the tomographic process is key to making learning-based approaches competitive with previously established methods.

3.4 Scaling to Three Qubits

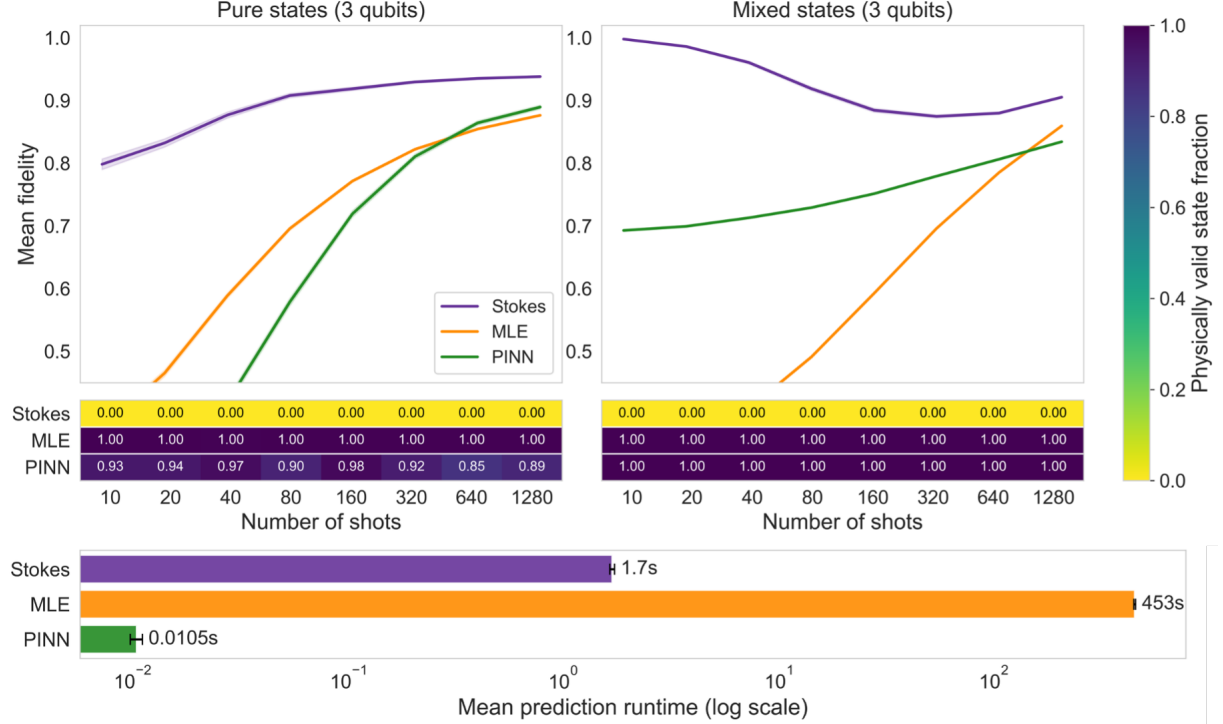


Figure 9: Extension of the reconstruction methods to three-qubit states. Top panels show mean reconstruction fidelity as a function of the number of measurement shots for pure and mixed states. Middle panels show the fraction of physically valid reconstructed states. Bottom panel shows mean reconstruction runtime on a logarithmic scale. The results highlight how the relative performance of Stokes reconstruction, MLE, and the PINN changes as the dimensionality of the system increases. The y-axis is cut for visibility at higher shot numbers.

To assess how the reconstruction methods extend beyond the two-qubit setting, the same pipeline was applied to three-qubit states, as shown in Fig. 9. This provides a more challenging test of performance, as the dimension of the state space increases exponentially with the number of qubits, placing greater demands on both the reconstruction algorithms and the available measurement data.

For pure states, Stokes again achieves the highest fidelities across all shot numbers, increasing from 0.80 to approximately 0.94. The PINN and MLE both start from much lower fidelities, below the cut-off y-axis, reflecting the increased difficulty of reconstruction in higher dimensions, but improve steadily with shot number. The neural network shows a particularly strong recovery, approaching the performance of MLE at high shot counts, but both remain slightly below Stokes. This behaviour suggests that while the learned mapping becomes more difficult to predict as system size increases, the MLE and the neural network are still able to reconstruct the states accurately given sufficient data. For mixed states, the differences between the methods are less pronounced, with all approaches improving gradually and approaching similar fidelities at higher shot numbers.

An interesting feature of Stokes reconstruction is the non-monotonic behaviour ob-

served at low to intermediate shot numbers, where the fidelity decreases from near unity before increasing again. This may be due to the fact that at very low shot counts, the estimated probabilities are highly coarse-grained, taking only a small number of discrete values. While statistical noise is large in this regime, small systematic noise arising from experimental imperfections, such as waveplate misalignment or reduced visibility, may not be resolved at this level of precision and therefore may not significantly affect the input frequencies that Stokes uses to reconstruct the state. As a result, the reconstruction can appear stable. As the number of shots increases, however, the probability estimates become sufficiently precise for these systematic deviations to be reflected in the data. When combined with the linear inversion in Stokes reconstruction, these structured errors can be amplified, leading to a temporary decrease in fidelity before the estimates become more accurate at higher shot counts. A similar effect was present in the two-qubit results shown in Sec. 3.3, although it was much less pronounced, likely due to the lower dimensionality of the system and the reduced impact of these structured errors.

The physicality results remain largely consistent with the two-qubit case. Stokes reconstruction continues to produce unphysical states across all regimes, while MLE enforces perfect physicality by construction. The PINN again achieves perfect physicality for mixed states and maintains a high fraction of valid states for pure states, demonstrating that the parametrisation remains robust even as the system size increases.

The most significant differences between the methods emerge in their computational scaling. The runtime for Stokes reconstruction increases substantially compared to the two-qubit case, and MLE becomes even more computationally demanding, reflecting the growing cost of solving a constrained optimisation problem in a higher-dimensional space. In contrast, the neural network shows only a minimal increase in inference time, remaining orders of magnitude faster than both classical methods.

Overall, these results suggest that while classical methods remain competitive in smaller systems, their limitations become more apparent as the system size increases. The PINN, although requiring more data to achieve comparable fidelity in the three-qubit setting, retains strong physical consistency and has much better scaling in computational cost[10]. This highlights its potential as a practical approach for QST in larger systems, where both reconstruction speed and physical validity are critical. However, it is worth noting that the experimental setup simulated in this work is for two entangled photons, so the three-qubit analysis is meant to illustrate how the methods scale, rather than to directly represent the experimental system considered here.

4 Limitations and Future Work

While the results presented demonstrate the potential of PINNs for QST, several limitations of the present study should be acknowledged. The analysis is restricted to two and three-qubit systems, which, although useful for benchmarking, do not fully capture the challenges associated with larger quantum systems. Additionally, all results are based on simulated data, meaning that real experimental effects such as calibration errors, drift, and imperfect measurement models are not fully represented[7]. As a result, the performance of the neural network in practical experimental settings may differ from that observed here[17].

There are several directions in which the methodology could be improved. The PINN’s architecture is informed by the physical structure of the tomography problem but still remains relatively simple, and further improvements could be achieved by incorporating

stronger features that more closely reflect the structure of the problem. This could include more tailored parametrisations, improved loss functions, or architectures designed to better capture correlations in the measurement data. In addition, while MLE was implemented with a fixed optimisation procedure, increasing the number of optimisation steps or improving convergence criteria could lead to more competitive performance, albeit at a significantly increased computational cost.

A key area for future work is the application of these methods to experimental datasets. This would allow for a more realistic assessment of performance, particularly in the presence of systematic errors and noise sources that are difficult to simulate accurately. Furthermore, extending the approach to larger qubit systems is an important next step. While classical methods are still competitive in small systems, the results presented here suggest that neural network approaches may offer greater advantages as the dimensionality increases, particularly in terms of computational scaling.

Finally, there is scope to include other machine learning methods into the reconstruction pipeline. For example, as shown by Palmieri et al.[4], neural networks can be used to minimise noise to give a new set of noise-reduced measurement probabilities. This two-step process could allow the reconstruction neural network to more accurately reconstruct the state, without also needing to account for noise within the reconstruction stage itself. Overall, while the current work provides a strong foundation, further development is required to fully realise the potential of neural network-based QST in practical and scalable settings.

5 Conclusion

This project set out to investigate the use of neural networks for QST, with a particular focus on whether incorporating physical structure into the model could improve reconstruction performance. Stokes reconstruction and Maximum Likelihood Estimation (MLE) were used as classical benchmarks, allowing for a direct comparison across fidelity, physicality, and computational efficiency.

The results show that the PINN provides a strong overall balance between these metrics. While Stokes achieves high fidelity, particularly in low-shot regimes, it does not guarantee physically valid states. MLE, on the other hand, guarantees physicality and improves with increasing measurement data, but is computationally expensive and scales poorly. The PINN does not outperform the classical methods in every regime, but in regimes with enough shots, it achieves competitive fidelity, maintains strong physicality, and provides extremely fast reconstruction once trained. Analysis of the fidelity distributions shows that the neural network produces more consistent reconstructions at higher shot numbers, while scaling to three-qubit systems highlights its favourable computational scaling and consistent physicality of reconstructed states compared to classical methods.

Overall, these results suggest that incorporating prior physical knowledge into neural network architectures is an important factor in making learning-based approaches viable for QST. While classical methods remain effective in small systems, the balance of performance and efficiency offered by the PINN makes it a promising method for QST.

References

- [1] Daniel F. V. James, Paul G. Kwiat, William J. Munro, and Andrew G. White. Measurement of qubits. *Physical Review A*, 64(5), 2001. doi:10.1103/physreva.64.052312.
- [2] Nengkun Yu. Sample efficient tomography via pauli measurements, 2020.
- [3] K. Aditi and Stephen Becker. Rigorous maximum-likelihood estimation for quantum states. *Physical Review A*, 112(5), 2025. doi:10.1103/j5gh-hmtw.
- [4] Adriano Macarone Palmieri, Egor Kovlakov, Federico Bianchi, Dmitry Yudin, Stanislav Straupe, Jacob D. Biamonte, and Sergei Kulik. Experimental neural network enhanced quantum tomography. *npj Quantum Information*, 6(1), 2020. doi:10.1038/s41534-020-0248-6.
- [5] Artur Czerwinski. Selected concepts of quantum state tomography. *Optics*, 3(3): 268–286, 2022. doi:10.3390/opt3030026.
- [6] Shawn X. Wang and Gregory S. Kanter. Robust multiwavelength all-fiber source of polarization-entangled photons with built-in analyzer alignment signal. *IEEE Journal of Selected Topics in Quantum Electronics*, 15(6):1733–1740, 2009. doi:10.1109/JSTQE.2009.2022278.
- [7] Emil R. Hellebek, Klaus Mølmer, and Anders S. Sørensen. Characterization of the multimode nature of single-photon sources based on spontaneous parametric down conversion. *Physical Review A*, 110:023728, 2024. doi:10.1103/PhysRevA.110.023728.
- [8] Shabnam Jabeen, Dmytro Kurdydyk, Aadi Palnitkar, Mihir Talati, Jeffrey Yan, and Jinghong Yang. Quantum machine learning for state tomography using classical data, 2025.
- [9] A. I. Lvovsky and M. G. Raymer. Continuous-variable optical quantum-state tomography. *Reviews of Modern Physics*, 81(1):299–332, 2009. doi:10.1103/revmodphys.81.299.
- [10] Giacomo Torlai, Guglielmo Mazzola, Juan Carrasquilla, Matthias Troyer, Roger G. Melko, and Giuseppe Carleo. Neural-network quantum state tomography. *Nature Physics*, 14(5):447–450, 2018. doi:10.1038/s41567-018-0048-5.
- [11] Midhun T. Augustine. A survey on universal approximation theorems, 2024.
- [12] Sanjaya Lohani, Brian T. Kirby, Michael Brodsky, Onur Danaci, and Ryan T. Glasser. Machine learning assisted quantum state estimation. *Machine Learning: Science and Technology*, 1(3):035007, 2020. doi:10.1088/2632-2153/ab9a21.
- [13] Fabio A. González, Alejandro Gallego, Santiago Toledo-Cortés, and Vladimir Vargas-Calderón. Learning with density matrices and random features. *Quantum Machine Intelligence*, 4(2), 2022. doi:10.1007/s42484-022-00079-9.
- [14] Karol Życzkowski and Hans-Jürgen Sommers. Hilbert–schmidt volume of the set of mixed quantum states. *Journal of Physics A: Mathematical and General*, 36(39): 10115–10130, 2003. doi:10.1088/0305-4470/36/39/310.
- [15] Edward A. West and Matthew H. Smith. Polarization errors associated with birefringent waveplates. In *Polarization Analysis and Measurement II*, volume 2265, pages 260–271, 1994. doi:10.1117/12.186698.
- [16] Benyamin Ghogh and Mark Crowley. The theory behind overfitting, cross validation, regularization, bagging, and boosting: Tutorial, 2023.
- [17] Matthias Christandl and Renato Renner. Reliable quantum state tomography. *Physical Review Letters*, 109(12):120403, 2012. doi:10.1103/PhysRevLett.109.120403.

Summary for a General Audience

Quantum computers encode information in “qubits.” Like bits, they can be 0 or 1, but a qubit can also exist in a superposition, effectively a blend of 0 and 1 at the same time. To build and test quantum hardware, researchers must precisely identify the states of these qubits. This process, called quantum state tomography, uses many measurements to predict the underlying quantum state. This is like trying to reconstruct a 3D object only from shadows cast from different directions.

As quantum devices grow in size, however, the data required to specify their states explodes. Standard reconstruction techniques either risk predicting “unphysical” states that break quantum rules, or they rely on time-consuming optimisation that must be redone for each new dataset.

This project investigated whether neural networks can learn to reconstruct quantum states from these measurements more effectively. The central idea was to encode the laws of quantum mechanics directly into the network’s architecture, in an attempt to improve performance and ensure that its outputs are always valid quantum states while it is trained on simulated experimental measurements.

The study found that this physics-aware neural network matches the accuracy of established methods, runs orders of magnitude faster after training, and scales more gracefully to larger numbers of qubits, pointing to a powerful route for characterising advanced quantum technologies.

Generative AI Statement

Generative AI was used in this report primarily to assist with the structure and formatting of the LaTeX document, and to help in troubleshooting errors in the developed code for the project. It was also used for developing figures, as well as spellchecking the final report and suggestions for improving structure/ readability. All content, analysis, and decision-making for the method pipeline was developed and verified by the author.